Chapter 21: Column, row, and nullspaces Worksheet

1. Are each of the following statements true or false?

(a) The rank of a matrix is always less than or equal to the minimum of the number of its rows and columns.

(b) The rank of the zero matrix is always 1.

(c) The rank of the sum of two matrices is always the sum of their ranks.

(d) If A is a 3×3 matrix with $\text{rank}(A) = 2$, then $\det(A) = 2$.

(e) If a matrix has a row of all zeros, its rank will be less than the total number of rows.

(f) If two matrices are row equivalent, they have the same rank.

(g) The rank of a matrix remains unchanged when the matrix is transposed.

2. Find a basis for $\text{col}(A)$, where

$$A = \begin{bmatrix} 1 & 3 & 0 & 3 & 2 \\ -1 & 0 & -9 & 9 & 4 \\ 1 & 3 & 0 & 5 & -2 \\ 2 & 4 & 6 & -3 & 2 \end{bmatrix}, \quad RREF(A) = \begin{bmatrix} 1 & 0 & 9 & 0 & -22 \\ 0 & 1 & -3 & 0 & 10 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

What's the dimension of $\text{col}(A)$? (Note: you computed $\text{null}(A)$ on a previous worksheet)

3.

(a) Can a 3×4 matrix have linearly independent columns? Linearly independent rows? Explain.

(b) If A is is 4×3 and $\text{rank}(A) = 2$, can A have linearly independent columns? Linearly independent rows? Explain.

(c) If A is an $m \times n$ matrix and $\text{rank}(A) = m$, show that $m \leq n$.

(d) Can a nonsquare matrix have its rows linearly independent and its columns linearly independent? Explain.

(e) Can the null space of a 3×6 matrix have dimension 2? Explain.

(f) Suppose that A is 5×4 and $\text{null}(A) = \{c\mathbf{x}\}$ for some column vector $\mathbf{x} \neq \mathbf{0}$. Can $\dim \text{col}(A) = 2$?